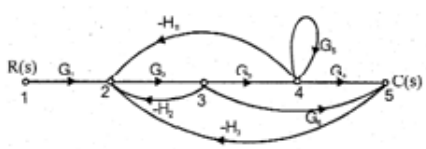


CONTROL SYSTEMS – UE24EE241B

Sample Questions

Signal Flow Graphs	
1.	Which of the following is NOT allowed in a forward path? A. Repetition of nodes B. Positive gain C. Negative gain D. Multiple branches
2.	Two loops are said to be non-touching if: A. They have same gain B. They do not share any node C. They have same direction D. They intersect once
3.	According to Mason's Gain Formula, the determinant Δ is: A. Sum of all loop gains B. Product of all loop gains C. $1 - (\text{sum of loop gains})$ D. $1 - (\text{sum of loop gains}) + (\text{sum of gain products of non-touching loops})$
4.	If a unit step response of a system is impulse function, then what is the transfer function of the system?
5.	In the signal flow graph of figure given below,  Find the following : 1. Number of forward path and its gain 2. loop gains 3. Δ 4. Transfer function
6.	For a system with one forward path and one loop touching the forward path , the overall gain is: A. $\frac{P}{1 - L}$ B. $\frac{P}{1 + L}$ C. $P(1 - L)$ D. $P(1 + L)$

7.	The value of Δ_k in Mason's formula represents: A. Total loop gain B. Forward path gain C. Determinant excluding loops touching the k-th forward path D. Product of all non-touching loops
8.	If two non-touching loops have gains L_1 and L_2, their contribution to Δ is: A. $+L_1L_2$ B. $-L_1L_2$ C. $+2L_1L_2$ D. $-2L_1L_2$
9.	In Mason's Gain Formula, the numerator is: A. $\sum P_k$ B. $\sum P_k \Delta$ C. $\sum P_k \Delta_k$ D. $\prod P_k$

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(Anchor)